

Cambridge (CIE) IGCSE Chemistry



Your notes

Group Properties & Trends

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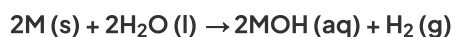
- * Group I Properties
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- * Transition Elements
- * Noble Gases

Chemical properties of the Group I metals



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- They react readily with oxygen and water vapour in air
 - This is why they are stored under **oil**, to stop them from reacting
- Group I metals will react similarly with water
 - They react to form an **alkaline** metal hydroxide solution and **hydrogen** gas

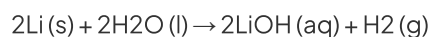


- The Group I metals get more reactive as you move down the group
 - Only the first three metals are allowed in schools for demonstrations

Reactions of the Group I metals and water

Lithium

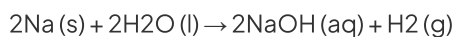
lithium + water $\rightarrow$ lithium hydroxide + hydrogen



- Observations:
 - Relatively slow reaction
 - Fizzing
 - Lithium moves on the surface of the water

Sodium

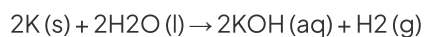
sodium + water $\rightarrow$ sodium hydroxide + hydrogen



- Observations:
 - More vigorous fizzing
 - Moves rapidly on the surface of the water
 - Dissolves quickly

Potassium

potassium + water $\rightarrow$ potassium hydroxide + hydrogen



- Observations:
 - Reacts more vigorously than sodium
 - Burns with a **lilac** flame
 - Moves very rapidly on the surface

- Dissolves very quickly

General trends of the Group I metals

- As you move down the group:
 - Melting point decreases
 - Density increases
 - There is an exception in the early part of the group
 - Reactivity (with water) increases

Predicting the Properties of Group I Elements

- Knowing the reactions of elements at the top of the group allows you to predict the properties of other elements further down Group I

Properties of other alkali metals

- The reactivity of alkali metals increases down the group
- This means that rubidium, caesium and francium will react more vigorously with air and water than lithium, sodium and potassium
 - Lithium will be the **least** reactive metal at the top of the group
 - Francium will be the **most** reactive at the bottom
- Francium is rare and radioactive so is difficult to confirm predictions
- For example the reactions with water can be predicted:

Predicting the reaction with water

Rubidium

- Predicted observations:
 - Very violent reaction
 - May explode with sparks

Caesium

- Predicted observations:
 - Extremely violent reaction
 - Can explode due to rapid production of heat and hydrogen

Francium

- Predicted observations:



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- Extremely reactive
- Difficult to confirm experimentally due to rarity and radioactivity

You can use any given data for Group 1 elements to identify trends and make predictions about other elements in the group.



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Group VII properties & trends

- These are the Group VII non-metals that are **poisonous** and include fluorine, chlorine, bromine, iodine and astatine
- Halogens are **diatomic**
 - This means that they form molecules containing **two** atoms
- The formulae of the halogens are:
 - Fluorine = F_2
 - Chlorine = Cl_2
 - Bromine = Br_2
 - Iodine = I_2
 - Astatine = At_2
- All halogens have seven electrons in their outer shell
 - They form **halide** ions by gaining one more electron to complete their outer shells
- Fluorine is not allowed in schools so observations and experiments tend to only involve chlorine, bromine and iodine

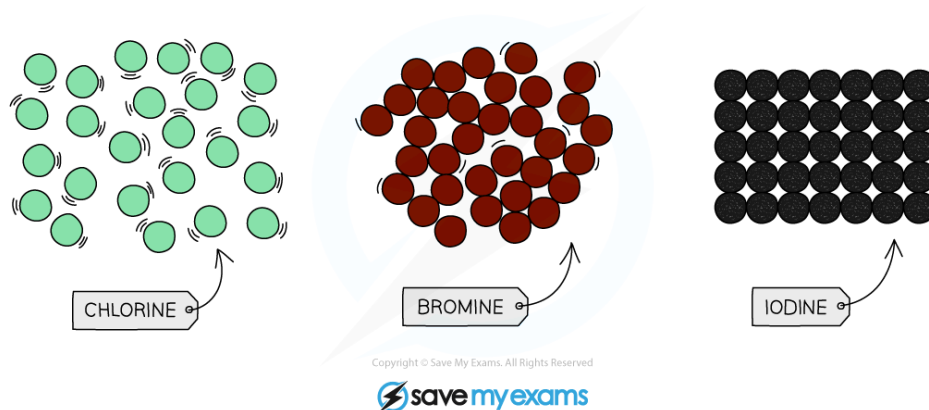
Properties of the halogens

Physical state, colour and density

- At room temperature (20 °C), the physical state of the halogens changes as you go down the group
 - Chlorine is a **pale yellow-green gas**
 - Bromine is a **red-brown liquid**
 - Iodine is a **grey-black solid**
- This demonstrates that the **density** of the halogens **increases** as you go **down the group**:



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The physical state of the halogens at room temperature



Examiner Tips and Tricks

Solid iodine, iodine in solution and iodine vapour are different colours.

- Solid iodine is dark **grey-black**
- Iodine vapour is **purple**
- Aqueous iodine is **brown**.

Reactivity

- The **reactivity** of Group 7 elements **decreases** as you go down the group
 - This is the opposite trend to Group 1
- When a halogen reacts, it needs to gain one outer electron to achieve a full outer shell of electrons
- As you go down Group 7, the number of electron shells **increases**:

Element	Number of shells	Electron configuration
Fluorine (F)	2	2, 7
Chlorine (Cl)	3	2, 8, 7
Bromine (Br)	4	Beyond the syllabus
Iodine (I)	5	Beyond the syllabus

- The increasing number of electron shells has two main effects:
 - The **atomic radius increases**, so the outer shell is further from the nucleus



- There is more **electron shielding** from the inner shells
- These factors reduce the force of attraction between the nucleus and an incoming electron
- Therefore, it becomes harder to attract an electron
- This means that the reactivity of the halogens **decreases** down the group

Predicting group VII properties

- You may be given information about some elements and asked to predict the properties of other elements in the group
- The information you might be given could be in relation to melting/boiling point or physical state/density so it is useful to know the trends in properties going down the group

Predicting melting and boiling point

- The melting and boiling point of the halogens **increases** as you go down the group
- Fluorine is at the top of Group 7
 - It has the **lowest** melting and boiling point
- Astatine is at the bottom of Group 7
 - It has the **highest** melting and boiling point

Predicting physical states and density

- The halogens become **denser** as you go down the group
- Fluorine is at the top of Group 7
 - It is a **gas**
- Astatine is at the bottom of Group 7
 - It is a **solid**

Predicting colour

- The colour of the halogens becomes **darker** as you go down the group
- Fluorine is at the top of Group 7 so the colour will be **lighter**
 - Fluorine is **yellow**
- Astatine is at the bottom of Group 7 so the colour will be **darker**
 - Astatine is **black**



Examiner Tips and Tricks

If you are doing the extended course you can be asked to identify trends in chemical or physical properties of the Group 7 elements, given appropriate data.

Firstly, make sure that you have placed the elements and associated data in either ascending or descending order according to their position in Group 7. Then look for any general patterns in the data.



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Group VII displacement reactions

- A halogen **displacement reaction** occurs when a more reactive halogen displaces a less reactive halogen from an aqueous solution of its halide
- The reactivity of Group 7 non-metals increases as you move up the group
- Out of the three commonly used halogens, chlorine, bromine and iodine, chlorine is the most reactive and iodine is the least reactive
 - This is because a more reactive halogen will oxidise the halide ions of a less reactive halogen:
 - The more reactive halogen gains electrons to form halide ions
 - The less reactive halide ions lose electrons to form the halogen element

Colour of halogens in aqueous solutions

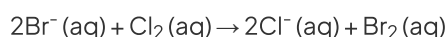
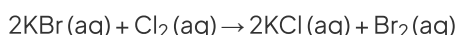
Aqueous solution	Colour
Chlorine	Very pale green but usually appears colourless as it is very dilute
Bromine	Orange but will turn yellow when diluted
Iodine	Brown

Halogen displacement reactions

Chlorine and bromine

- If you add chlorine solution to colourless potassium bromide solution, the solution becomes orange as bromine is formed
- Chlorine is **above** bromine in Group 7 so is more reactive
- Chlorine will therefore **displace** bromine from an aqueous solution of the metal bromide
- The less reactive halogen is displaced and forms the halogen element

potassium bromide + chlorine → potassium chloride + bromine

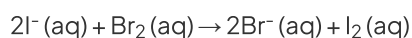
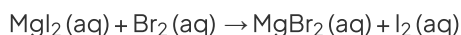


Bromine and iodine

- Bromine is above iodine in Group 7 so is **more** reactive

- Bromine will therefore **displace** iodine from an aqueous solution of metal iodide
- The solution will turn brown as iodine is formed

magnesium iodide + bromine → magnesium bromide + iodine



Summary table of displacement reactions

	Chlorine (Cl ₂)	Bromine (Br ₂)	Iodine (I ₂)
Potassium chloride (KCl)	x	No reaction	No reaction
Potassium bromide (KBr)	Chlorine displaces the bromide ions Yellow-orange colour of bromine seen	x	No reaction
Potassium iodide (KI)	Chlorine displaces the iodide ions Brown colour of iodine is seen	Bromine displaces the iodide ions Brown colour of iodine is seen	x



Examiner Tips and Tricks

Iodine solid, solution and vapour are different colours. Solid iodine is dark **grey-black**, iodine vapour is **purple** and aqueous iodine is **brown**.



Your notes

Transition elements

- Transition elements are found in the centre of the [Periodic Table](#) between Group 2 and Group 3
- Most metals are transition elements and have properties typical of a metal:
 - They are **lustrous**
 - They are **hard** and **strong**
 - They **conduct** heat and electricity
 - They have **high melting points**
 - Except mercury: This has a low melting point which is why it is a liquid at room temperature

The transition elements in the Periodic Table

1		2												3		4		5		6		7		8
Li		Be		TRANSITION METALS										B		C		N		O		F		Ne
Na		Mg												Al		Si		P		S		Cl		Ar
K		Ca		Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr					
Rb		Sr		Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe					
Cs		Ba		La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn					
Fr		Ra		Ac																				



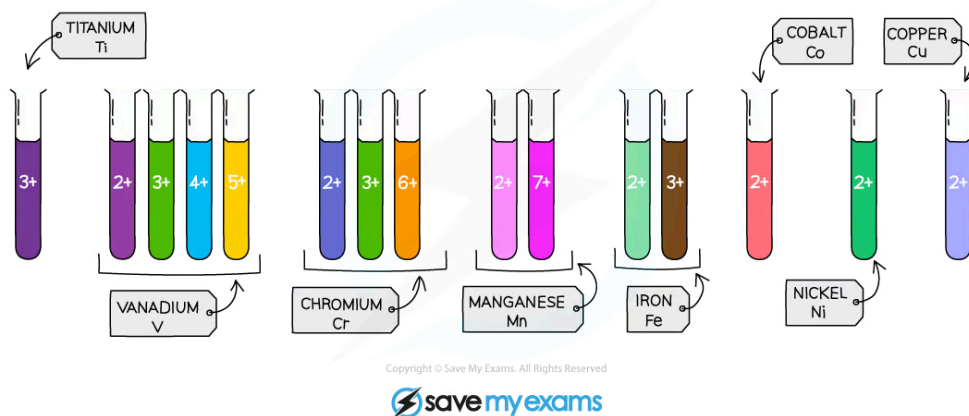
Transition elements are found in the middle of the Periodic Table

- There are some properties that are only typical of transition elements :
 - They form **ions with different charges**
 - E.g. Iron forms Fe^{2+} and Fe^{3+} ions
 - They are useful as **catalysts**
 - E.g. Iron is used as the catalyst in the [Haber process](#)
- They form **coloured compounds**
 - E.g. Copper forms blue copper sulfate, black copper oxide and green copper carbonate

Coloured compounds formed by transition metal ions



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The colours produced by some transition elements ions



Examiner Tips and Tricks

Although scandium and zinc are in the transition element area of the Periodic Table, they are not considered transition elements as they do **not** form coloured compounds and have only **one** oxidation state.

Transition elements oxidation numbers

Extended tier only

- The transition elements have more than one **oxidation number**, as they can lose a different number of electrons
- For example, iron either:
 - Lose **two** electrons to form Fe^{2+} so has an oxidation number of +2
 - Loses **three** electrons to form Fe^{3+} so has an oxidation number of +3
- Compounds containing transition elements in different oxidation states will have different **properties** and colours



Examiner Tips and Tricks

Transition elements also referred to as transition metals.

- | | | | | | | | | | | | | | | | | | |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------------|----|
| | | | | | | | | | | | | | | | | NOBLE GASES | |
| | | | | | | | | | | | | | | | | 0 | |
| 1 | 2 | | | | | | | | | | | 3 | 4 | 5 | 6 | 7 | He |
| Li | Be | | | | | | | | | | | B | C | N | O | F | Ne |
| Na | Mg | | | | | | | | | | | Al | Si | P | S | Cl | Ar |
| K | Ca | Sc | Ti | V | Cr | Mn | Fe | Co | Ni | Cu | Zn | Ga | Ge | As | Se | Br | Kr |
| Rb | Sr | Y | Zr | Nb | Mo | Tc | Ru | Rh | Pd | Ag | Cd | In | Sn | Sb | Te | I | Xe |
| Cs | Ba | La | Hf | Ta | W | Re | Os | Ir | Pt | Au | Hg | Tl | Pb | Bi | Po | At | Rn |
| Fr | Ra | Ac | | | | | | | | | | | | | | | |



- The noble gases:
 - Are **non-metals**
 - Are **monoatomic**
 - This means that they are made of a single atom
 - Are **colourless**
 - Are **gaseous**
 - Have very **low melting** and **boiling** points
 - Are unreactive / **inert**

Why the noble gases are inert

- The Group VIII noble gases have **full outer shells**
- Electronic configurations of the noble gases:
 - He: 2
 - Ne: 2,8
 - Ar: 2,8,8



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- Kr: 2,8,18,8
- Xe: 2,8,18,18,8
- This means that the noble gases have no need to gain or lose electrons
- These electronic configurations are **extremely stable**
- As a result, noble gases do not easily form bonds with other elements